

EDUCATION

University of Pennsylvania <i>M.S.E. in Robotics, GPA: ≥ 4</i> Birla Institute of Technology and Science, Pilani (BITS Pilani) <i>B.E. in Mechanical Engineering — Minor in Data Science, CGPA: 8.57/10</i> Institute Merit-cum-Need Scholarship — Consecutive Recipient (Top 8% of Students)	Pennsylvania, USA <i>Aug '26 – May '28</i> Rajasthan, India <i>Oct '22 – May '26</i>
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PUBLICATIONS & PEER REVIEWS

Journal Publications R. Chakraborty and K. Kishore, “<i>Navigating Without Satellites: A Critical Survey of GNSS-Denied UAV Navigation.</i>” Submitted to IEEE Transactions on Artificial Intelligence (T-AI), 2026. 🔗 R. Chakraborty and K. Kishore, “<i>Multi-Critic Off-Policy Reinforcement Learning for Quadrotor Velocity Control in Unknown Indoor Environments.</i>” Manuscript in preparation for IEEE Transactions on Artificial Intelligence (T-AI), 2026. Conference Publications Y. Wang, H. He, Y. Cao, J. Liang, R. Chakraborty, and G. Sartoretti, “<i>CogniPlan: Uncertainty-Guided Path Planning with Conditional Generative Layout Prediction.</i>” Accepted at Conference on Robot Learning (CoRL), 2025; published in Proceedings of Machine Learning Research (PMLR). 🔗 R. Chakraborty, T. Mian, and P. Kundu, “<i>An Efficient Approach for Synthetic Data Generation and Fault Diagnosis for Rotating Machinery.</i>” Accepted at Prognostics and System Health Management Conference (PHM), 2025; published in IET Conference Proceedings. 🔗 K. Kishore and R. Chakraborty, “<i>Path Planning of Low-Altitude UAV for Tree Canopy Tracking and Orchard Monitoring.</i>” Publication awaiting intellectual property clearance, 2025. 🔗 Poster Presentations R. Chakraborty, “<i>High-Velocity Impact Testing and Analysis of Layered Metal-Composite Shields Using Electromagnetic Railgun.</i>” Presented as poster at Pulsed Power Science, Technology and Application (PPSTA), 2024. Peer Reviews “<i>GS-SDE: 3D Gaussian Splatting-based Adaptive Scene Modeling in Semi-Dynamic Environments.</i>” Peer review completed for IEEE Transactions on Automation Science and Engineering (T-ASE), 2026. “<i>Limits of Wearable Gait Analysis for Age Classification in Young Adults: Anthropometry Over Age.</i>” Peer review completed for IEEE India Council International Subsections Conference (INDISCON), 2026. “<i>Generative AI and Computer Vision Based Multi-Assessment System for Early Prediction and Diagnosis of Autism Spectrum Disorder (ASD).</i>” Peer review completed for IEEE India Council International Subsections Conference (INDISCON), 2026. “<i>AquaNode: Design and Implementation of a Low-Cost Automated Water Tank Monitoring and Valve Control System Using Arduino and Ultrasonic Sensing.</i>” Peer review completed for IEEE India Council International Subsections Conference (INDISCON), 2026. “<i>A research paper on IoT-Based Smart Irrigation Systems.</i>” Peer review completed for IEEE India Council International Subsections Conference (INDISCON), 2026. “<i>Hybrid AI-Based Autonomous Drone Swarm Navigation.</i>” Peer review completed for IEEE Gujarat Section Conference (GSC), 2026. “<i>Autonomous Robot Navigation for Smart Indoor using A* and Deep Q-Network Model.</i>” Peer review completed for IEEE Gujarat Section Conference (GSC), 2026. “<i>Deep Learning Driven Robot Simulation for Autonomous Navigation, Crop Health and Pesticide Spraying.</i>” Peer review completed for IEEE Gujarat Section Conference (GSC), 2026.	
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EXPERIENCE – ROBOTICS & COMPUTER VISION

Vision-Language Grounded Multi-Robot Coordination and Navigation <i>Supervisor: Dr. Avinash Gautam, Associate Professor, BITS Pilani</i> - Surveyed literature on vision-language grounded multi-robot coordination and VLMs-based spatial representations. - Reviewed decentralized navigation strategies fusing 3D visual-language maps with LLM-driven instruction parsing.	<i>May '25 – Aug '25</i>
Off-Policy Multi-Critic RL with Reward Decomposition for Indoor Exploration 🔗 <i>Supervisor: Dr. Kaushal Kishore, Principal Scientist, CSIR-CEERI</i> - Integrated three specialized critics for forward progress, lateral avoidance, and angular alignment with distinct reward signals. - Deployed multi-critic TD3 with prioritized replay achieving 94.2% success rate and 22% lower collision rate versus baseline.	<i>Jan '26 – May '26</i>
Vision-Attention-Driven Autonomous Navigation with Semantic Understanding <i>Supervisor: Dr. Guillaume Sartoretti, Assistant Professor, MARMoT Lab, NUS</i> - Enhanced CogniPlan with cross-attention between frontier and node embeddings to enrich node representations. - Incorporated Visual Navigation Transformer (ViNT) to capture semantic context for adaptive exploration strategies.	<i>Aug '25 – Dec '25</i>
Design and Development of Smart Automated Field Hockey Ball Launcher 🔗 <i>Supervisor: Dr. Mani Shankar Dasgupta, Associate Professor, BITS Pilani</i> Physics-Based Launcher Mechanical Design and FEM Analysis - Fabricated programmable launcher achieving 150 km/h using CFRP composite and stainless steel components. - Performed FEM analysis and CAD using Fusion 360, achieving safety factor 2.07 at 496 RPM. Computer Vision Pipeline for 3D Trajectory Reconstruction and Shot Classification - Devised 3D ball localization from monocular videos using diameter-based depth and PnLCalib (0.76m RMSE). - Constructed TCN-attention classifier fusing trajectory and statistical features, achieving 96.4% classification accuracy.	<i>Apr '24 – Oct '25</i>
Uncertainty-Guided Path Planning via Conditional Layout Prediction 🔗 <i>Supervisor: Dr. Guillaume Sartoretti, Assistant Professor, MARMoT Lab, NUS</i> - Adapted multiple image-based exploration planners into ROS pipelines for evaluation in the CMU Exploration Environment. - Supported the development and benchmarking of CogniPlan, achieving up to 17.7% shorter exploration paths and 3.9% improved navigation efficiency over state-of-the-art baselines across 100+ unseen maps.	<i>Mar '25 – Aug '25</i>
Monocular Vision-Based UAV Navigation for Orchard Monitoring 🔗 <i>Supervisor: Dr. Kaushal Kishore, Principal Scientist, CSIR-CEERI</i> - Engineered a UAV-based orchard monitoring system using YOLOv11 (Box mAP50: 95.5%, Mask mAP50: 96.5%). - Implemented B-spline trajectory logic and coded custom yaw-roll controller to minimize drift under wind.	<i>Jan '24 – Feb '25</i>
Image-Based Visual Servoing for Automated Radar Control and UAV Tracking <i>Supervisor: Dr. Rishi Verma, Head (ESDS), PPEMD, BARC</i> - Established automated radar alignment using YOLOv8 and DeepSORT, improving tracking recall to 91% and speed by 13%. - Interfaced a PLC-based actuation system enabling continuous UAV lock-in within $\pm 5^\circ$ angular error.	<i>Jun '24 – Jul '24</i>

EXPERIENCE – GENERATIVE AI & PREDICTIVE MAINTENANCE

Synthetic Data Generation for Bearing Fault Diagnosis Under Data Scarcity <i>Supervisor:</i> Dr. Pradeep Kundu , Assistant Professor, KU Leuven <i>Auxiliary Classifier</i> WGAN-GP for Time-Series Sensor Data Generation ↗ - Built ACWGAN-GP with TCN discriminator for minority fault class augmentation reaching ~100% classification accuracy. - Assessed synthetic data using PCC, Cosine Similarity, MMD, and KL Divergence on FFT spectra. <i>Physics-Informed Generative Modeling for Bearing Fault Diagnosis Under Data Scarcity</i> - Formulated physics-informed cVAE-GAN and cVQ-GAN frameworks isolating latent-space effects in CWT scalogram synthesis. - Validated LiteFormer classification gains across Density, Coverage, and CMMD metrics.	<i>Sep '24 – May '26</i>
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EXPERIENCE – MECHANICAL SIMULATION & BEHAVIORAL RESEARCH

Finite Element Analysis of Running Track Surfaces <i>Supervisor:</i> Dr. Mani Shankar Dasgupta , Associate Professor, BITS Pilani - Modeled surface materials using hyperelastic formulations, evaluating restitution across 3 sports disciplines. - Proposed optimized track designs improving durability by 23% based on ANSYS fatigue analysis. Primal World Beliefs in Engineering Students ↗ <i>Supervisor:</i> Dr. Rajneesh Choubisa , BITS Pilani - Conducted a primal world beliefs study surveying engineering undergraduates on core worldview dimensions. - Replicated established population-level findings, validating trends against the original general-population study. Impact Testing and Analysis of Metal-Composite Shields Using Electromagnetic Railguns <i>Supervisor:</i> Dr. Rishi Verma , Head (ESDS), PPEMD, BARC - Characterized 3 material configurations via the Johnson-Cook model for high-strain-rate impact response. - Verified simulation results through railgun tests at 3.2 MJ energy, achieving <7% deviation from FEM predictions.	<i>Apr '24 – Sep '25</i> <i>Aug '24 – Dec '24</i> <i>Jun '24 – Jul '24</i>
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INDUSTRY EXPERIENCE

ML Intern Uber (AI Solutions) <i>Supervisor:</i> Mr. Siddarth Malreddy , Tech Lead Manager & Mr. Ishan Nigam , Senior ML Engineer, Uber - Streamlined human-in-the-loop automated object tracking in uLabel for RGB/IR videos. - Flagged object annotation anomalies via XGBoost (mAP50: 75%). - Screened object-association errors with a TrackTrack-based cross-frame consistency check (F1: 88%). Subject Matter Expert Knowledge Cultivator <i>Self-supervised</i> - Resolved numerical problems for student clients using MATLAB, CAD, and Python. - Authored academic content across diverse research topics.	<i>Jul '25 – Dec '25</i> <i>Nov '23 – Feb '24</i>
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PROJECTS

Vision-Guided Robotic Arm for Simulated Oral Cancer Screening - Simulated a KUKA KR 150 R3100 arm in MATLAB/Simulink for end-effector guidance inside a modeled oral cavity. - Trained an EfficientNet-B0 model to detect cancerous regions, localizing the end effector through ORB-SLAM3 tracking. Autonomous Drone Navigation MathWorks Global Student Drone Challenge 2025 - Designed vision-based control for Parrot Mambo drone using masking, ray-tracing, and closed-loop yaw control. - Optimized speed control and zone-based auto-landing to reduce track completion time. Air Quality Index Forecasting Using Time Series Models ↗ - Benchmarked AR, MA, ARMA, ARIMA, and SARIMA models forecasting daily and weekly AQI for Jaipur and Kolkata. - Diagnosed stationarity with ADF and KPSS tests, tuning lag orders via ACF/PACF and scoring on R², MAE, and RMSE. 3D Lunar Surface Modeling from DEM Data ISRO Immersion Challenge 2024 ↗ - Created lunar terrain models from TMC DEM data with curvature extraction, rendered live in UE5. - Automated data import and heightmap transformation via a custom plugin, processing each tile in under 10s. Cricket Team Optimization AMEX Campus Challenge 2024 (Product Track) ↗ - Processed and cleaned 100K+ datapoints to identify anomalies, outliers, and consistency trends. - Generated optimal lineups using statistical heuristics, achieving a 14% gain over baseline strategies. ExoMy Rover Navigation and UR3 Arm Motion Planning ERC 2023 Remote ↗ - Navigated ExoMy rover using ArUco detection, Ackermann steering, and spot turns for autonomous hazard avoidance. - Calibrated UR3 arm with MoveIt and OMPL planner for collision-free manipulation (98% success).	<i>Aug '25 – Dec '25</i> <i>Mar '25 – Apr '25</i> <i>Nov '24 – Nov '24</i> <i>Jul '24 – Jul '24</i> <i>Jun '24 – Jul '24</i> <i>Apr '23 – Sep '23</i>
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AWARDS & ACHIEVEMENTS

Finalist — Meta PyTorch OpenEnv Hackathon (India) ↗	<i>Apr '26</i>
3rd Place — MathWorks Global Drone Student Challenge 2025	<i>Mar '25</i>
Finalist — ISRO Immersion Challenge (AI for Space & Geospatial Innovation) ↗	<i>Jul '24</i>
Top 15 Overall, Top 5 in College — American Express Campus Challenge 2024 ↗	<i>Jul '24</i>
5th Place & Best Maintenance Award — European Rover Challenge 2023 (Remote Finals)	<i>Sep '23</i>

TEACHING

Teaching Assistant <i>CIS 4190/5190: Applied Machine Learning, University of Pennsylvania</i> <i>CS F320: Foundations of Data Science, BITS Pilani</i> <i>ME F218: Advanced Mechanics of Solids, BITS Pilani</i> <i>ME F216: Materials Science and Engineering, BITS Pilani</i> - Assisted 100+ students in labs and tutorials, clarifying concepts and linking theory to practical applications. - Analyzed assignments and facilitated faculty in delivering high-impact teaching sessions.	<i>Aug '26 – Dec '26</i> <i>Jan '26 – May '26</i> <i>Jan '25 – May '25</i> <i>Sep '24 – Dec '24</i>
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LEADERSHIP & VOLUNTEERING

Deputy Director of Academic Programming <i>Graduate and Professional Student Assembly (GAPSA), University of Pennsylvania</i>	<i>Aug '26 – Present</i>
Vice-President <i>South Asian Association (Rangoli), University of Pennsylvania</i>	<i>Aug '26 – Present</i>
President & Secretary <i>Mechanical Engineering Association (MEA), BITS Pilani</i> - Coordinated 10+ events and career sessions for 300+ students, facilitating technical exposure and alumni interaction. - Managed production of 500+ merchandise items and led outreach, boosting student participation by 20%.	<i>Jun '24 – May '26</i>

President & Tech Fest Coordinator	
<i>Indian Society of Heating, Refrigerating, and Air Conditioning Engineers (ISHRAE), BITS Pilani</i>	
- Led a team of 25+ members to organize 4+ technical workshops with HVAC industry experts. - Hosted 3 competitions and networking events, engaging 200+ students in HVAC innovation and awareness.	
Project Manager	
<i>Tinkerer's Lab (TL), BITS Pilani</i>	
- Supervised 5 interdisciplinary robotics teams (30+ members) on Micromouse and Hexapod projects. - Oversaw lab resources, conducted weekly reviews, and mentored 50+ students in hands-on technical skills.	
Treasurer	
<i>Bengali Cultural Association (Moruchhaya), BITS Pilani</i>	
Allocated funds and maintained financial records for merchandise sales, cultural programs, and annual events.	
Volunteer	
<i>Junoon, National Service Scheme (NSS), BITS Pilani</i>	
Organized sports and recreational competitions for specially abled children from 25+ colleges.	

PROFESSIONAL ACTIVITIES

Graduate Student Member — IEEE	<i>2026</i>
COMPETITIVE EXAMINATIONS	
GRE: 317 (Quantitative: 168, Verbal: 149, AWA: 3)	<i>2025</i>
TOEFL iBT: 108 (Reading: 27, Listening: 30, Speaking: 23, Writing: 28)	<i>2025</i>
JEE Advanced: All India Rank 14372	<i>2022</i>

TECHNICAL SKILLS & COURSEWORK

Relevant Coursework: Machine Learning, Deep Learning, Machine Perception, Computer Vision, Foundations of Data Science, Applied Statistical Methods, Linear Algebra, Engineering Optimization, Differential Equations, Design of Machine Elements, Digital Twins
Programming Languages: Python, C++, C, C#, Java, Bash (Shell)
Robotics & Simulation: ROS (with Gazebo, Rviz), MAVROS, Navigation Stack, MoveIt, AirSim, Unity, Unreal Engine, MATLAB, Simulink, QGIS
AI & Data Science: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, OpenCV, Open3D, NumPy, Pandas, SciPy, StatsModels, Matplotlib, Seaborn, Weights & Biases (W&B), Excel
Cloud & MLOps: AWS (EC2, SageMaker), Azure (Foundry), Google Cloud Platform (GCP), Docker, Git, GitHub Actions
Hardware & Embedded Systems: NVIDIA Jetson (Nano, Orin), Raspberry Pi, Arduino, IMUs, Stereo Camera, 3D LiDAR
CAD & Mechanical Simulation: ANSYS Mechanical, SolidWorks, Fusion 360
Leadership & Communication: Event Planning, Team Management, Budget Allocation, Mentorship, Cross-Functional Coordination, Public Speaking, Stakeholder Outreach, Technical Writing
Productivity Software: Microsoft Office (Excel, PowerPoint, Word), Google Workspace